BIOCON MALAYSIA

MASTER PROOF OF APPROVED PRINTED **CARTRIDGE LEAFLET** ART WORK

Insugen®-N 100 IU/mL, 3 mL FOR COLOUR COPY

(FOR DUOPHARMA - MALAYSIA MFG.) ANNEXURE- 4 [Ref. No. : BM/QA/SOP/059]

ARTWORK CODE: **BM/XXX/XX**

For the use of only a registered medical practitioner or hospital or laboratory

Biocon

DUOPHARMA

Insugen®- N (NPH)

Insulin, Human of recombinant DNA Origin. For s.c. use only

100 IU/mL

3 mL cartridge

Insulin Injection, Isophane

NAME OF THE MEDICINAL PRODUCT

Insugen®-N (NPH)
(Insulin Injection, Isophane 100 IU/mL)

COMPOSITION

Each ml contains:

Insulin, Human Ph. Eur.	100 IU
(Insulin,Human of recombinant DNA Origin)	
m-Cresol Ph. Eur.	0.16% w/v
Phenol Ph. Eur.	0.065% w/v
Water for injecton Ph. Eur.	q.s

One IU (International Unit) of insulin is equivalent to 0.035 mg of human insulin. Each 3 mL cartridge contains suspension for injection equivalent to 300 IU.
For a full list of excipients, see section **List of Excipients.**
Insulin human, rDNA is produced by recombinant DNA technology in *Pichia pastoris.*

PHARMACEUTICAL FORM

Suspension for injection in a cartridge.
A white suspension which on standing deposits white sediment and leaves a colourless or almost colourless supernatant liquid, the sediment is readily resuspended by gentle shaking.

PHARMACOLOGICAL PROPERTIES

Pharmacotherapeutic group: insulins and analogues for injection. Intermediate-acting, insulin (human). ATC Code: A10AC01
INSUGEN®-N has been developed as a similar biological medicinal product to Insulatard®.

Mechanism of Action

The blood glucose lowering effect of insulin is due to the facilitated uptake of glucose following binding of insulin to receptors on muscle and fat cells and to the simultaneous inhibition of glucose output from the liver.

Insugen®-N (NPH) is a human insulin with longer duration of action as compared to regular human insulin.

Onset of action is within 1.5 hours, reaches a maximum effect within 4 to 12 hours and the entire duration of action is approximately 24 hours.

Pharmacodynamic properties:

The blood glucose lowering effect of insulin is due to the facilitated uptake of glucose following binding of insulin to receptors on muscle and fat cells and to the simultaneous inhibition of glucose output from the liver. Insulin in the blood stream has a half-life of a few minutes. Consequently, the time-action profile of an insulin preparation is determined solely by its absorption characteristics. This process is influenced by several factors (e.g. insulin dosage, injection route and site), which is why considerable intra and inter patient variants are seen.

Clinical efficacy of INSUGEN®-R plus INSUGEN®-N was evaluated in comparison with Actrapid® plus Insulatard® in a randomized, open-label, two-phase, 48-week, phase 3 study in type 1 diabetes mellitus (T1DM) patients (Study INSUGCT300509). During the first 24-week comparative phase, patients received INSUGEN®-R plus INSUGEN®-N or Actrapid® plus Insulatard® whereas, in the subsequent 24-week non-comparative phase, patients who received Actrapid® plus Insulatard® were switched to receive INSUGEN®-R plus INSUGEN®-N. During the study, efficacy of INSUGEN®-R plus INSUGEN®-N was similar to Actrapid® plus Insulatard® with respect to changes in glycosylated haemoglobin for standardization (HbA1C), fasting plasma glucose, 7-point capillary blood glucose levels and prandial as well as basal insulin dose. Switching the patients from Actrapid® plus Insulatard® to ®INSUGEN-R plus INSUGEN-N had no effect on efficacy. Overall, INSUGEN®-R and INSUGEN®-N were individually similar to Actrapid® and Insulatard® in efficacy.

Pharmacokinetic Properties

Insulin in the blood stream has a half-life of a few minutes. Consequently, the time-action profile of an insulin preparation is determined solely by its absorption characteristics.

This process is influenced by several factors (e.g. insulin dosage, injection route and site thickness of subcutaneous fat, type of diabetes). The pharmacokinetics of insulin products are therefore affected by significant intra-and inter-individual variation.

Absorption

The maximum plasma concentration of insulin is reached within 2 to 18 hours after subcutaneous administration.

Distribution

No profound binding to plasma proteins, except circulating insulin antibodies (if present) has been observed.

Metabolism

Human Insulin is reported to be degraded by insulin protease or insulin-degrading enzymes and possibly protein disulphide isomerase. A number of cleavage (hydrolysis) sites on the human insulin molecule have been proposed; none of the metabolites-formed following the cleavage are active.

Elimination

The terminal half-life (t1/2) is determined by the rate of absorption from the subcutaneous tissue. The t1/2 is therefore a measure of the absorption rather than of the elimination per se of insulin from plasma (insulin in the blood stream has a t½ of a few minutes). Trials have indicated a t1/2 of about 5 to 10 hours.

Preclinical Safety Data

Non-Clinical data reveal no special hazard for humans based on conventional studies of safety pharmacology, repeated dose toxicity, genotoxicity, carcinogenic potential, and toxicity to reproduction.

CLINICAL PARTICULARS

Therapeutic indications

Treatment of diabetes mellitus.

Posology and Method of Administration

Insugen®-N (NPH) is a long-acting insulin.

Dosage

Dosage is individual and determined in accordance with the needs of the patient. The individual insulin requirement is usually between 0.3 and 1.0 IU/kg/day. The daily insulin requirement may be higher in patients with insulin resistance (e.g., during puberty or due to obesity) and lower in patients with residual, endogenous insulin production.

The physician determines whether a single or several daily injections are necessary. Insugen®-N (NPH) may be used alone or mixed with fast-acting insulin. In intensive insulin therapy the suspension may be used as basal insulin (evening and/or morning injection) with fast-acting insulin given at meals.

In patients with diabetes mellitus, optimised glycaemic control delays the onset of late diabetic complications. Close blood glucose monitoring is therefore recommended.

Dosage Adjustment

Concomitant illness, especially infections and feverish conditions usually increases the patient's insulin requirement.

Renal or hepatic impairment may reduce insulin requirement.

Adjustment of dosage may also be necessary if patients change physical activity or their usual diet. Dosage adjustment may be necessary when transferring patients from one insulin preparation to another (see section **Special Warnings and Precautions for Use**).

Administration

Insulin suspensions are for subcutaneous use and never to be administered intravenously.

Insugen®-N (NPH) is administered subcutaneously in the thigh or abdominal wall. If convenient, the gluteal region or the deltoid region may also be used.

Subcutaneous injection into the abdominal wall ensures a faster absorption than from other injection sites.

Injection into a lifted skin fold minimises the risk of unintended intramuscular injection.

The needle should be kept under the skin for at least 10 seconds to make sure the entire dose is injected. If blood appears after the needle has been withdrawn, press the injection site lightly with a finger. Injection sites should always be rotated within the same region in order to reduce the risk of lipodystrophy and cutaneous amyloidosis.

Contraindications

Hypersensitivity to the active substance or to any of the recipients (see section **List of Excipients**).
Hypoglycaemia.

Special Warnings and Precautions for Use

Before travelling between different time zones, the patient should be advised to consult the physician, as the patient may require taking insulin and meals at different times.

Hyperglycaemia:

Inadequate dosage or discontinuation of treatment, especially in type 1 diabetes, may lead to hyperglycaemia. Usually, the first symptoms of hyperglycaemia set in gradually, over a period of hours or days. They include thirst, increased frequency of urination, nausea, vomiting, drowsiness, flushed dry skin, dry mouth, loss of appetite as well as acetone odour of breath. In type 1 diabetes, untreated hyperglycaemic events eventually lead to diabetic ketoacidosis, which is potentially lethal.

Hypoglycaemia:

Hypoglycaemia may occur if the insulin dose is too high in relation to the insulin requirement (see sections Side Effects and Overdose). Omission of a meal or unplanned, strenuous physical exercise may lead to hypoglycaemia. Patients, whose blood glucose control is greatly improved by intensified insulin therapy, may experience a change in their usual warning symptoms of hypoglycaemia and should be advised accordingly. Usual warning symptoms may disappear in patients with longstanding diabetes. Concomitant illness, especially infections and feverish conditions, usually increases the patient's insulin requirement. Concomitant diseases in the kidney, liver or affecting the adrenal, pituitary or thyroid gland can require changes in the insulin dose.

Transfer from other Insulin Medicinal Products:

Transferring a patient to another type or brand of insulin should be done under strict medical supervision. Changes in strength, brand (manufacturer), type (fast-, dual-,long-acting insulin etc.), origin (animal, human or analogue insulin) and/or method of manufacture (recombinant DNA versus animal source insulin) may result in a need for a change in dosage. If an adjustment is needed when switching the patients to INSUGEN®-N, it may occur with the first dose or during the first several weeks or months.

When patients are transferred between different types of insulin medicinal products, the early warning symptoms of hypoglycaemia may change or become less pronounced than those experienced with their previous insulin.

A few patients who have experienced hypoglycaemic reactions after transfer from animal source insulin have reported that early warning symptoms of hypoglycaemia were less pronounced or different from those experienced with their previous insulin.

Injection Site Reactions:

As with any insulin therapy, injection site reactions may occur and include pain, itching, hives, swelling and inflammation. Continuous rotation of the injection site within a given area may help to reduce or prevent these reactions. Reactions usually resolve in a few days to a few weeks. On rare occasions, injection site reactions may require discontinuation of INSUGEN®-N. Due to the risk of precipitation in pump catheters, INSUGEN®-N should not be used in insulin pumps for continuous subcutaneous insulin infusion.

Skin and subcutaneous tissue disorders

Patients must be instructed to perform continuous rotation of the injection site to reduce the risk of developing lipodystrophy and cutaneous amyloidosis. There is a potential risk of delayed insulin absorption and worsened glycaemic control following insulin injections at sites with these reactions. A sudden change in the injection site to an unaffected area has been reported to result in hypoglycaemia. Blood glucose monitoring is recommended after the change in the injection site and dose adjustment of antidiabetic medications may be considered.

Combination of INSUGEN®-N with Pioglitazone:

Cases of cardiac failure have been reported when pioglitazone was used in combination with insulin, especially in patients with risk factors for development of cardiac heart failure. This should be kept in mind if treatment with the combination of pioglitazone and INSUGEN®-N is considered. If the combination is used, patients should be observed for signs and symptoms of heart failure, weight gain and oedema. Pioglitazone should be discontinued if any deterioration in cardiac symptoms occurs.

Drug interactions

A number of medicinal products are known to interact with glucose metabolism. The physician must therefore take possible interactions into account and should always ask his patient about any medicinal products they take.

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Insulin, Human of recombinant DNA Origin. For s.c. use only

100 IU/mL

3 mL cartridge

Insulin Injection, Isophane

The following substances may reduce insulin requirement:

Oral hypoglycaemic agent (OHA), monoamine oxidase inhibitors (MAO), non-selective beta-blocking agents, angiotensin converting enzyme. (ACE) inhibitors, salicylates, alcohol, anabolic steroids and sulphonamides

The following substances may increase insulin requirement:

Oral contraceptives, thiazides, glucocorticoids, thyroid hormones and beta-sympathomimetics., growth hormone and danazol.

Beta-blocking agents may mask the symptoms of hypoglycaemia and delay recovery from hypoglycaemia.

Octreotide/lanreotide may both decrease and increase insulin requirement.

Alcohol may intensify and prolong the hypoglycaemic effect of insulin.

Pregnancy and Lactation

There are no restrictions on treatment of diabetes with insulin during pregnancy, as insulin does not cross the placental barrier.

Both hypoglycaemia and hyperglycaemia, which can occur in inadequately controlled diabetes therapy, increase the risk of malformations and death in *utero*. Intensive control in the treatment of pregnant women with diabetes is therefore recommended throughout pregnancy and when contemplating pregnancy.

Insulin requirements usually fall in the first trimester and subsequently increase during the second and third trimesters. After delivery, Insulin requirements return rapidly to pre-pregnancy values.

Insulin treatment of the nursing mother presents no risk to the baby. However, the Insugen®-N (NPH) dosage may need to be adjusted.

Effects on Ability to Drive and Use Machines

The patient's ability to concentrate and react may be impaired as a result of hypoglycaemia. This may constitute a risk in situations where these abilities are of special importance (eg, driving a car or operating machinery).

Patients should be advised to take precautions to avoid hypoglycaemia while driving. This is particularly important in those who have reduced or absent awareness of the warning signs of hypoglycaemia or have frequent episodes of hypoglycaemia. The advisability of driving should be considered in these circumstances.

Side Effects:

As for other insulin products, in general, hypoglycaemia is the most frequently occurring undesirable effect. It may occur if the insulin dose is too high in relation to the insulin requirement. In clinical trials and during marketed use, the frequency varies with patient population and dose regimens. Therefore, no specific frequency can be presented. Severe hypoglycaemia may lead to unconsciousness and/or convulsions and may result in temporary or permanent impairment of brain function or even death.

Safety data for INSUGEN®-R plus INSUGEN®-N was generated in a randomized, open-label, two-phase, 48-week, phase 3 study in T1DM patients (Study INSUGCT300509). Incidence of treatment-emergent adverse events (TEAEs), severe adverse events and serious adverse events were similar for INSUGEN®-R plus INSUGEN®-N and Actrapid® plus Insulatard® during the study. There were no notable clinically significant differences when patients switched from Actrapid® plus Insulatard® to INSUGEN®-R plus INSUGEN®-N.

The TEAEs occurring in ≥2% of patients in either treatment group during the study are described below:
General disorders and administration site conditions: asthenia and pyrexia
Nervous system disorders: headache
Infections and infestations: nasopharyngitis, influenza, tonsillitis, upper respiratory tract infection and urinary tract infection
Skin and subcutaneous tissue disorders: hyperhidrosis
Musculoskeletal and connective tissue disorders: back pain and pain in extremity
Metabolism and nutrition disorders: dyslipidemia, hypercholesterolemia and hypoglycaemia
Injury, poisoning and procedural complications: excoriation
Renal and urinary disorders: ketonuria and proteinuria
Respiratory, thoracic and mediastinal disorders: cough
Vascular disorders: hypertension

The following side effects and their frequencies have been observed under treatment with Insugen®-N. The assessment of side effects is based on the following frequency data: uncommon (≥1/1,000 to <1/100). Isolated spontaneous cases are presented as very rare defined as <1/10,000 including isolated reports.

Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

Nervous system disorders
Uncommon - Peripheral neuropathy
Fast improvement in blood glucose control may be associated with a condition termed "acute painful neuropathy", which is usually reversible.

Eye disorders
Very rare side effect - Refraction disorders
Refraction anomalies may occur upon initiation of insulin therapy. These symptoms are usually of transitory nature.
Uncommon side effect - Diabetic retinopathy
Long-term improved glycaemic control decreases the risk of progression of diabetic retinopathy. However, intensification of insulin therapy with abrupt improvement in glycaemic control may be associated with temporary worsening of diabetic retinopathy.

Skin and subcutaneous tissue disorders
Uncommon - Lipodystrophy
Not known - Cutaneous amyloidosis
Lipodystrophy and cutaneous amyloidosis may occur at the injection site and delay local insulin absorption. Continuous rotation of the injection site within the given injection area may help to reduce or prevent these reactions.

General disorders and administration site conditions
Uncommon - Injection site reactions
Injection site reactions (redness, swelling, itching, pain and haematoma at the injection site) may occur

during treatment with insulin. Most reactions are transitory and disappear during continued treatment.
Uncommon - Oedema
Oedema may occur upon initiation of insulin therapy. These symptoms are usually of transitory nature.

Immune system disorders
Uncommon - Urticaria, rash.
Very rare - Anaphylactic reactions
Symptoms of generalised hypersensitivity may include generalised skin rash, itching, sweating, gastrointestinal upset, and angioneurotic oedema, difficulties in breathing, palpitation, reduction in blood pressure and fainting /loss of consciousness. Generalised hypersensitivity reactions are potentially life threatening.

Overdose

A specific overdose of insulin cannot be defined. However, hypoglycaemia may develop over sequential stages:

- Mild hypoglycaemic episodes can be treated by oral administration of glucose or sugary product. It is therefore recommended that the diabetic patients carry some sugar lumps, sweet, biscuits or sugary fruit juice.
- Severe hypoglycaemic episodes, where the patient has become unconscious, can be treated by glucagon (0.5 to 1 mg) given intramuscularly or subcutaneously by a person who has received appropriate instruction, or by glucose given intravenously by a medical professional. Glucose must also be given intravenously, if the patient does not respond to glucagon within 10 to 15 minutes.

Upon regaining consciousness, administration of oral carbohydrate is recommended for the patient in order to prevent relapse.

PHARMACEUTICALS PARTICULARS

List of Excipients

Glycerol, metacresol, hydrochloric acid, sodium hydroxide, protamine sulphate, zinc oxide, phenol, dibasic sodium phosphate and water for injection.

Incompatibilities

Insulin product should only be added to compounds with which it is known to be compatible. Insulin suspensions should not be added to infusion fluids.

Shelf Life: Refer to label/ carton.

Interchangeability and Automatic Substitution

INSUGEN®-N has been developed as a similar biological medicinal product to Insulatard® and has been shown to have a comparable quality, safety and efficacy profile to Insulatard®. Therefore, interchange-ability with the reference product Insulatard® may be considered if this is in agreement with the treating physician. However, automatic substitution (i.e. the practice by which a different product to that specified on the prescription is dispensed to the patient without the prior informed consent of the treating physician) and active substance-based prescription cannot apply to biologicals, including biosimilars. Such a differentiating approach towards biologicals ensures that treating physicians can make informed decision about treatments is in the interest of patient's safety.

Storage and Precautions

Store in a refrigerator (2°C-8°C).
Do not store Insugen®-N (NPH) in or too near the freezer or cooling element
Do not freeze.
In-use Instructions: Do not refrigerate. Do not freeze Insugen®-N (NPH) cartridges that are in use can be kept at a temperature (not above 30°C) for up to 42 days.
Keep the cartridge in the outer carton in order to protect from light
Protect from excessive heat and direct sunlight.

Keep Insugen®-N (NPH) out of reach of children.
Jauhi daripada kanak-kanak.

Special Precautions for Disposal and Other Handling

Insulin preparations which have been frozen must not be used.
After removing Insugen®-N (NPH) cartridge from the refrigerator it is recommended to allow the cartridge to reach room temperature (not above 30°C) before resuspending the insulin as instructed for first time use.

Never use INSUGEN ® - N , if they do not appear uniformly white and cloudy after resuspension.

Any unused product or waste material should be disposed in accordance with local requirement.

Nature and content of container:
3 mL colourless tubular glass cartridges (Type 1) sealed using lined seals plugged with plunger stoppers and glass bead.

Pack Sizes: Currently available in 5x3 mL pack only

Direction Use of Cartridges.

For use with INSUPen EZ® and INSUPen Pro® (re-usable Injector) only

Please refer to User Manual for more information.

Product Registration Holder and Manufactured by:
Biocon Sdn. Bhd.
No.1, Jalan Bioteknologi 1,
Kawasan Perindustrian SiLC,
79200 Iskandar Puteri, Johor, Malaysia.

Marketed by:
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Revision Date: **May 2022**

BM/XXX/XX

PREPARED BY:	CHECKED BY:
DATE :	DATE :

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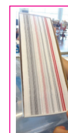
Size : 170 x 230mm (±1mm)
Paper : From India Supply: 45 GSM ITC newsprint paper (40 GSM - 50 GSM)
From Malaysia Supply: 50 GSM Woodfree Paper (45 GSM - 55 GSM)
Leaflet Folding Size : 170 x 29mm (±2mm)
Pharma Code : 4100
Folding pattern to be same as per attached dwg. no. 70M1163101

Colours:
PANTONE 541 C
PANTONE 375 C
BLACK C
PANTONE 2035 C
PANTONE 7624 C



1. Leaflets should not be bundled. Leaflets should be supplied in flat trays as per attached image.
2. Pharmacodes should be always on the outside for ensuring the reader to sense it.
3. The leaflet folding size should be same for all the leaflets.
4. Folded leaflets should not be reversed and inverted conditions.